

Finance-Neutral Output Gap Estimate and Policy Applications for Mexico^{1, 2}

1980–2025

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June 2026

¹ Master's thesis for the M.A. in Economics, El Colegio de México (2024–2026 cohort).

² The opinions and results presented are exclusively the author's and do not represent the position of the Bank of Mexico.

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Motivation

The 2008 crisis changed what we ask of the output gap

The Global Financial Crisis exposed a gap in macroeconomic diagnostics. A financial boom can sustain real output above its sustainable level and remain invisible to indicators built only on real activity. When the boom reverses, the contraction is severe, and the policy stance that preceded it looks indefensible in retrospect.

The lesson was that financial conditions are not separate from the real economy: they shape what counts as sustainable.

After 2008, treating “real” and “financial” cycles as independent is no longer defensible.

Mexico has lived this twice

For Mexico this is concrete experience, not theory. Before the 1994 peso crisis, real credit was growing above 20% per year, the peso was sustaining real appreciation, and equity prices had multiplied tenfold over six years—all signs of a financial boom that conventional gap estimates failed to flag. The crisis was severe: a 6.2% GDP contraction, depreciation above 50%, and a banking-system rescue costing roughly 15% of GDP.

The pattern repeated before 2008, this time with external financial conditions accumulating risk that domestic credit aggregates did not show. Both contractions are among the most severe in the country's postwar macroeconomic history.

Mexico's two largest contractions in the last forty years were preceded by visible financial booms that standard diagnostics did not capture.

No Mexican institutional estimate incorporates the financial cycle

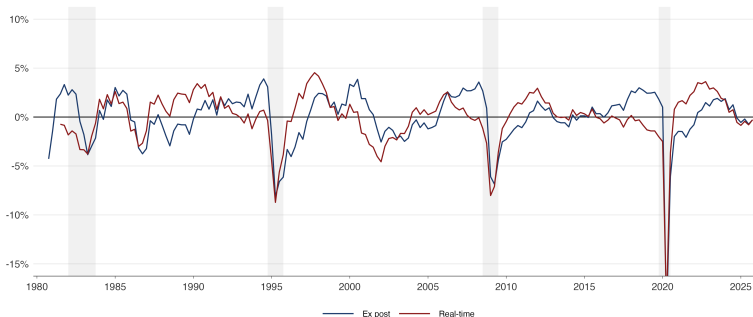
Banco de México publishes output-gap estimates derived from production-function and HP-type filters. The Secretaría de Hacienda uses HP-based cyclical adjustments in its fiscal accounting. The IMF Article IV country desk applies similar methods.

None of these incorporates credit, asset prices, the real exchange rate, or any other financial-conditions variable as a determinant of the gap.

Mexican policy is calibrated with diagnostics that, by construction, cannot see the kind of risk that drove 1994 and 2008.

HP also has a real-time problem

Set the conceptual issue aside for a moment. The HP filter, the workhorse of those institutional estimates, has a well-known end-of-sample bias: in real time it cannot see what the smoothed ex post series later reveals. In Mexico, the divergence is empirically large and concentrated precisely around the three contractions.



In all three episodes, ex post estimates later revealed the cyclical position; real-time HP did not.

The literature has a fix; for Mexico it has not been tested rigorously

Borio et al. (2013) proposed embedding financial variables directly into the state-space estimation of potential output—the *finance-neutral* gap. The approach worked for advanced economies.

Amador-Torres et al. (2016) extended it to three Latin American economies including Mexico, but with sample ending in 2013, only one variable beyond Borio's original three (the real exchange rate), and no deep analysis of Mexico or its policy implications.

There is a gap to close: a careful estimate of the finance-neutral gap for Mexico, with current data, the right financial variables, and explicit policy applications.

Question and contribution

Question. Can incorporating financial variables into output-gap estimation produce a more reliable signal of overheating for Mexico, particularly before crises?

Five contributions for Mexico:

1. Longest sample (45 years) ever applied with this methodology to Mexico.
2. Six financial variables organized in *domestic* and *external* blocks; equities, housing and commodities included for the first time in this framework.
3. Fully Bayesian estimation via adaptive MCMC (Haario et al., 2001)—credibility intervals, not point estimates.
4. Rigorous real-time evaluation with expanding window, including an explicit **pre-crisis window analysis**.
5. Three policy applications: fiscal, monetary, macroprudential.

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Methodology and data

Methodology: state-space, extending Borio et al. (2013)

To embed financial variables in the gap, we extend the state-space representation of the HP filter. The transition equation is preserved; only the observation equation is modified.

HP filter in state-space form.

$$\begin{aligned}\Delta y_t^* &= \Delta y_{t-1}^* + \varepsilon_{0t}, & \varepsilon_{0t} &\sim \mathcal{N}(0, \sigma_0^2) \quad (\text{transition}) \\ y_t &= y_t^* + \varepsilon_{1t}, & \varepsilon_{1t} &\sim \mathcal{N}(0, \sigma_1^2) \quad (\text{observation})\end{aligned}$$

With $\lambda_1 = \sigma_0^2/\sigma_1^2 = 1,600$ this is numerically identical to the standard HP filter.

Finance-neutral extension (Borio et al., 2013; Amador-Torres et al., 2016). Transition equation preserved; the observation equation is rewritten to embed an autoregressive gap and a vector of financial variables:

$$y_t - y_t^* = \beta (y_{t-1} - y_{t-1}^*) + \gamma' \mathbf{x}_{t-k*} + \varepsilon_{2t}, \quad \varepsilon_{2t} \sim \mathcal{N}(0, \sigma_2^2)$$

with $\beta \in (0, 0.95)$ and $\gamma_j \in (0, 1)$. If $\beta = 0$ and $\gamma = \mathbf{0}$, the model collapses to standard HP. Estimated by Kalman filter, the maximum-*a-posteriori* (MAP) and the adaptive Markov Chain Monte Carlo (MCMC) sampler of Haario et al. (2001).

Data: six financial variables, 1980Q1–2025Q4

Quarterly data from INEGI, Banxico, BIS and IMF. Variables enter in growth rates after Cesàro demeaning.

Domestic block	External block
Real credit (BIS)	Real multilateral exchange rate (BIS)
Equity prices (BMV / IPC)	Commodity export prices (Banxico)
Real housing price (BIS)	
Ex-post real rate (Banxico, INEGI)	

The sample spans three contraction episodes of distinct nature: **1994** (credit-driven), **2008–09** (externally transmitted), **2020** (real shock). The 2020 episode is the natural **placebo**: a contraction not preceded by financial overheating, against which to test whether the FN model over-attributes contractions to financial variables.

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Results

Estimation results

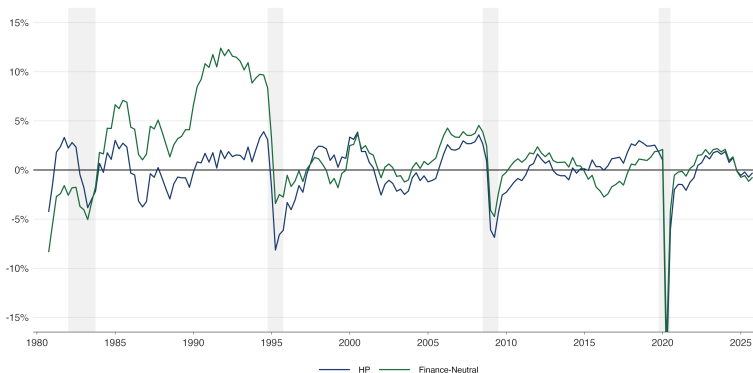
What did the data tell us about each channel? The table reports the posterior mean and posterior mean / sd ratio for persistence and financial-channel coefficients. The final specification keeps the five variables that meet the economic-significance criterion ($\bar{\gamma}_j \geq 0.001$); the ex-post real rate is excluded ($\bar{\gamma}_{dr} \approx 0.0005$, 60–80 $\times$ smaller than the rest).

Parameter	Posterior mean	mean / sd	k^*
β (persistence)	0.853	11.30	—
γ_{cr} — credit	0.0337	1.51	2
γ_{ap} — equities	0.0099	1.54	0
γ_{pv} — housing	0.0292	1.23	4
γ_{dq} — real FX	0.0489	2.48	0
γ_{ctot} — commodities	0.0351	1.69	0

All retained coefficients positive; posteriors concentrate away from zero. Real FX has the largest contribution per unit shock.

The finance-neutral gap for Mexico

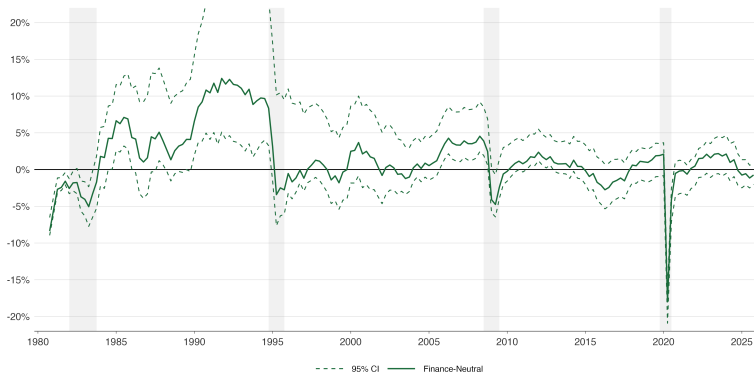
With the full sample, the FN gap tracks the HP filter for most of the period ($r = 0.579$) but diverges in magnitude around the two domestic crises. Between 1988 and 1994, FN reports overheating above 10% versus 3% for HP—consistent with the severity of the 1995 crisis.



Note the 2020 episode: because the nature of the crisis was sanitary, not financial, the FN gap converges with HP.

Credibility intervals around the FN gap

Bayesian estimation produces a full posterior distribution for the gap, not a point estimate. For each of the 2,500 retained MCMC draws, the Rauch–Tung–Striebel smoother is applied and a complete gap trajectory is computed. The 2.5 and 97.5 percentiles of the resulting distribution define the 95% credibility interval at every quarter.

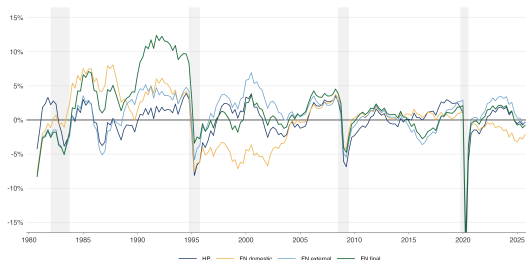


Quantifying gap uncertainty via credibility intervals—absent from Mexican institutional estimates—was

Which financial channels matter for Mexico?

Two findings stand out, both anchored in prior literature. Borio et al. (2013) found credit to be the dominant channel in advanced economies; Amador-Torres et al. (2016) showed the real FX dominates in LatAm including Mexico. I test both with a broader variable set and a domestic/external decomposition.

- Credit channel lower ($\gamma_{cr} = 0.034$)
confirms Borio for Mexico (low banking penetration)
- Real FX dominant ($\gamma_q = 0.049$)
confirms Amador-Torres with extended sample and richer specification
- Equities and commodities enter significantly; housing modest
- Domestic block more volatile; external block smoother but persistent



Broader specification reinforces prior findings; block decomposition reveals episode-specific dynamics.

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Real-time evaluation

Real-time performance: weak on average, strong before crises

For each quarter t from 1990Q1 to 2025Q4, the **entire model is recomputed using only the data available at t** : lag selection, Cesàro demeaning, and parameter estimation. No future information enters at any stage. This produces a real-time gap series of 144 observations to compare against the ex post estimate.

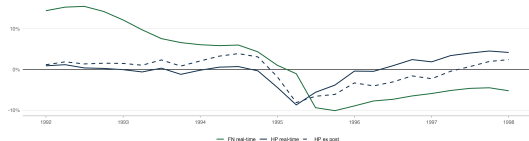
Metric	FN	HP
Full window, correlation r	0.694	0.707
Pre-crisis pooled ($n = 24$), correlation r	0.856	0.210

On the full window, FN does not improve upon HP on average—replicating the finding Amador-Torres et al. (2016) reported for Colombia. **Restricting evaluation to the eight quarters preceding each contraction** (1994, 2008, 2020) reverses the verdict: FN's correlation with the ex post diagnostic jumps to 0.856, while HP's collapses to 0.210.

The aggregate replicates Amador-Torres for Colombia. The pre-crisis decomposition is this thesis's contribution—and it reverses the verdict.

Crisis zoom: financial overheating in 1994 and 2008

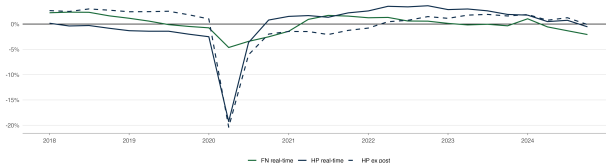
Three lines per panel: real-time HP (what policymakers saw), ex post HP (revealed later), real-time FN (what financial information signaled).



FN's reading before the event matches what ex post HP reveals years later.

Placebo test: the 2020 pandemic as a non-financial shock

The 2020 pandemic is the **placebo test** for the FN model: a real, non-financial shock with economic and financial consequences, but not preceded by a financial boom. If the model were biased toward financial attribution, it would still produce a large FN gap during the contraction. Instead, the FN gap converges with HP—recognizing the non-financial origin of the shock.



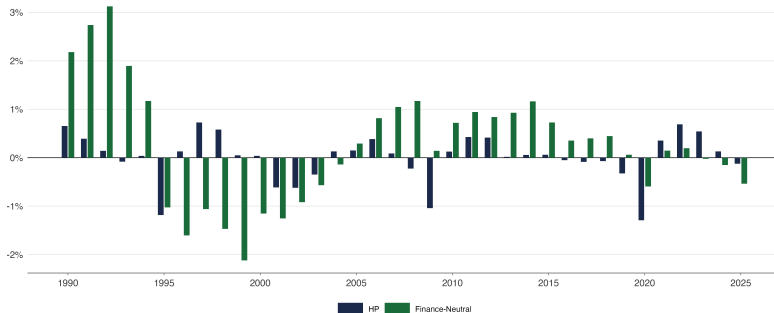
The 2020 episode serves as a placebo: a non-financial contraction against which to test whether FN over-attributes to financial variables. It does not.

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Policy applications

Application 1 — cyclically-adjusted fiscal balance

The cyclically-adjusted budget balance (CABB) separates the structural fiscal position from the part driven by the cycle—it is the standard OECD/IMF tool for assessing whether public finances are sustainable beyond the swings of activity. The decomposition is $b_t = b_t^* + \varepsilon \cdot \hat{g}_t$, so the cyclical correction is proportional to the gap; the gap choice directly shapes the diagnosis of structural balance. For Mexico, $\varepsilon = 0.21$ (Price et al., 2015).

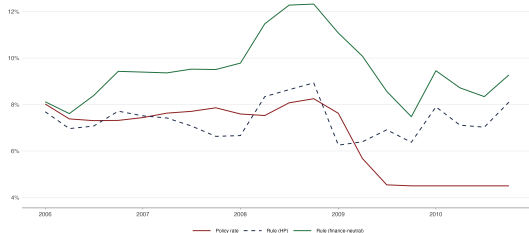


FN reveals weaker structural fiscal positions during financial booms.

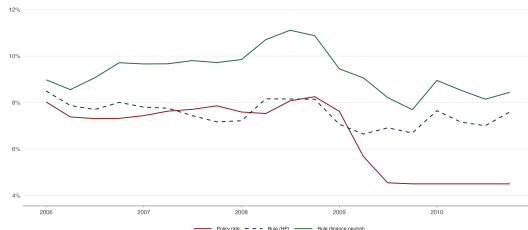
Application 2 — Taylor rules (GFC episode)

Two specifications, both evaluated in real time with only the gap measure differing: (i) Taylor (1993) calibrated with the original coefficients ($\phi_\pi = 1.5$, $\phi_y = 0.5$), and (ii) an OLS-estimated rule for Mexico's inflation-targeting regime, where the estimated weight on the output gap is small.¹

Taylor (1993) calibrated



Contemporary OLS rule for Mexico



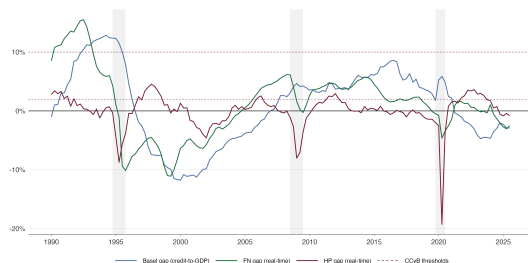
Under both Taylor specifications, FN prescribes a more prudent policy rate during financial booms.

¹The GFC is the only financial episode that falls inside the inflation-targeting regime (2003–): 1994 predates it and 2020 is non-financial.

Application 3 — countercyclical capital buffer (CCyB)

Basel III activates the CCyB when the credit/GDP gap crosses 2 p.p. FN evaluated as an alternative early-warning indicator against the Basel benchmark.

Indicator	Lead (q)		Detection (94 / 08)	False alarm
	1994	2008		
Basel credit/GDP	12	6	100% / 50%	44.0%
FN (real-time)	12	12	91.7% / 91.7%	33.9%
HP (real-time)	0	12	0% / 8.3%	24.8%



FN complements Basel's credit gap with better lead time and lower false-alarm rate—and identifies which channel drives the signal, pointing the macroprudential authority to the right instrument.

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Conclusions

What FN buys you

- ▶ Detects financial overheating HP misses (1994, 2008)
- ▶ Identifies the *channel* of risk (FX, equities, commodities, housing, credit)
- ▶ Provides a stable early-warning signal with consistent lead time
- ▶ Yields more conservative fiscal and monetary diagnostics during booms

Where it does not help

- ▶ Average-period stability is not improved over HP
- ▶ Performance comparable to HP for pure real shocks (2020)
- ▶ Long-horizon imbalances (>8 – 10 years): $\lambda = 1,600$ may understate

Limitations and next steps

Limitation. $\lambda = 1,600$ anchors the estimation to a business-cycle horizon, while financial cycles typically run 16–20 years (Drehmann et al., 2012). The model may capture only the shorter end of the financial-imbalance accumulation.

Next step. True real-time evaluation using INEGI's published vintages, replacing the current quasi-real-time exercise (which uses today's data without future quarters) with the data central bankers actually observed at each point in time. This is the priority robustness check for the paper.

Thank you

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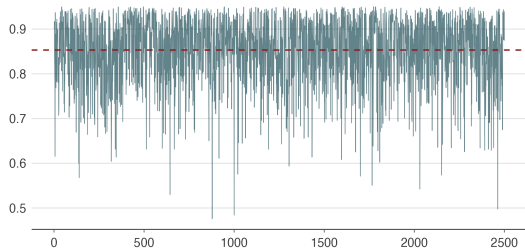
Appendix

Backup slides

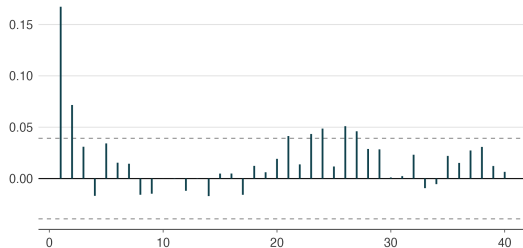
B1 — MCMC convergence

Adaptive Metropolis-Hastings (Haario et al., 2001), 50,000 draws, 20,000 burn-in, thinning 5.
Diagnostics shown for β (cycle persistence); behavior is representative across all parameters.

Trace plot — β



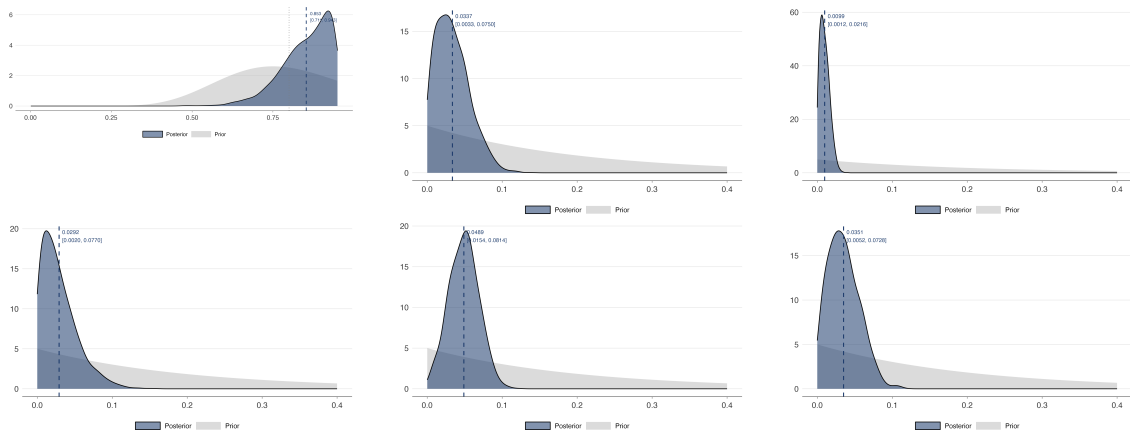
Autocorrelation — β



Chains mix; effective sample size adequate for all parameters.

B2 — Posterior distributions

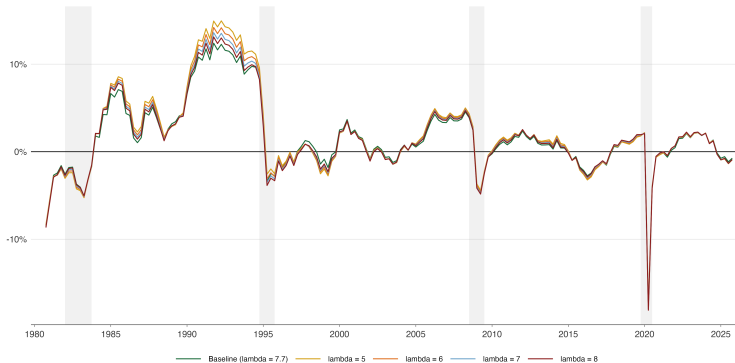
Posteriors vs. priors for β and all five retained γ_j coefficients.



Data are informative; posteriors concentrate away from priors.

B3 — Sensitivity to λ

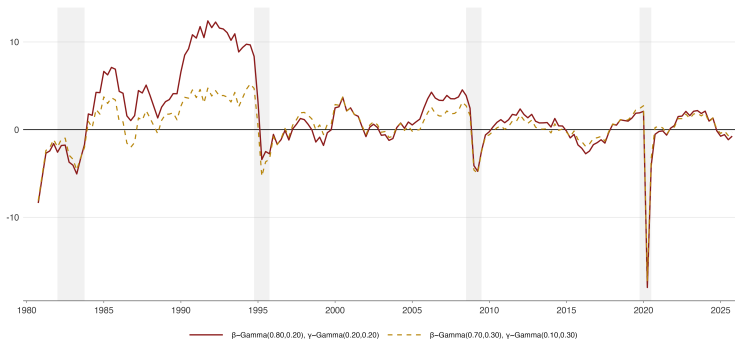
Estimates re-computed for $\lambda \in [800, 3,200]$.



Correlation with baseline > 0.994 for all λ considered.

B4 — Sensitivity to prior specification

Estimates re-computed under tighter and looser prior dispersion.



Results stable under reasonable variations in prior dispersion.

B5 — Parameter stability in real time

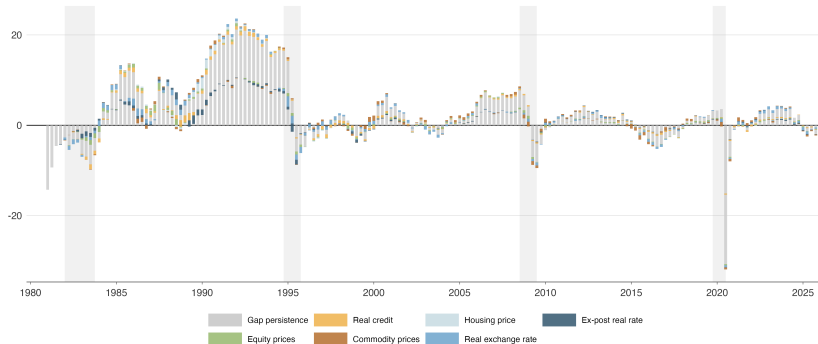
Mean, minimum and maximum of each posterior MAP across the 144 expanding-window estimations (1990Q1–2025Q4).

Param	Description	Mean RT	Min RT	Max RT	Ref. EP
β	Cycle persistence	0.8736	0.7834	0.9225	0.8531
γ_{dcr}	Credit	0.0719	0.0466	0.0937	0.0337
γ_{dap}	Equities	0.0133	0.0100	0.0158	0.0099
γ_{dpv}	Housing	0.0636	0.0332	0.1013	0.0292
γ_{dq}	Real FX	0.0392	0.0338	0.0514	0.0489
γ_{ctot}	Commodities	0.0423	0.0355	0.0561	0.0351

β never crosses 0.93; coefficients stable within reasonable bands across all 144 windows.

B6 — Channel-by-channel impulse

Per-variable contribution to the FN gap and cycle persistence.



1988–94 boom dominated jointly by credit, equities, housing, real FX; 2008 by external channels; 2020 mostly persistence.